Federated Learning With Soft Clustering

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***Abstract*—In this article, we consider the problem of federated learning (FL) with training data that are non independent and identically distributed (non-IID) across the clients. To cope with data heterogeneity, an iterative federated clustering algorithm (IFCA) has been proposed. IFCA partitions the clients into a number of clusters and lets the clients in the same cluster optimize a shared model. However, in IFCA, the clusters are nonoverlapping, which leads to an inefficient utilization of the local information since the knowledge of a client is used by only one cluster during each round. To capture the complex nature of real-world data, soft clustering methods with overlapping clusters have been proposed that attain superior performance over the hard ones. Motivated by this, we propose a new algorithm named FL with soft clustering (FLSC) by combining the strengths of soft clustering and IFCA, where the clients are partitioned into overlapping clusters and the information of each participating client is used by multiple clusters simultaneously during each round. The experimental results show that FLSC achieves better learning performance on the classification tasks on the MNIST and Fashion-MNIST data sets, compared with the state-of-the-art baseline methods, i.e., the global model method and IFCA.**

***Index Terms*—Data heterogeneity, federated learning (FL), soft clustering.**

I. Introduction

THE Internet of Things (IoT) devices, such as smartphones and sensors [3] are currently being widely used [2], [34], [36]. These generate a huge volume of data that contribute to the popularity of data-driven solutions to various practical problems, such as image classification [4], [35] and spectrum sensing [1]. Generally, these data-driven methods are centralized, which means that the central server collects decentralized data from different IoT clients in order to train machine learning (ML) models, such as deep neural networks. Although attaining satisfactory learning performance, the centralized ML approaches lead to many practical concerns, including the heavy communication burden induced by the transmission of raw data sets and the violation of data privacy of the clients [5], [6].

To address the above concerns while still benefiting from the large volume of decentralized training data, federated learning (FL) has been proposed as a decentralized alternative to the general centralized ML framework, where the data generated by the local clients are kept decentralized [5]–[12]. Basically, a shared global model is learned, which aggregates the information from the local clients as shown in the representative FL framework in Fig. 1 [8]. During each round, the central server broadcasts the global model to all the participating clients in the current round, and then each participating client trains the model with its own data set and sends the updated model to the server. The server aggregates them by averaging the received models. Hereinafter, as done in [17], we refer to the FL algorithm that learns a shared global model as shown in Fig. 1 as the global model method.

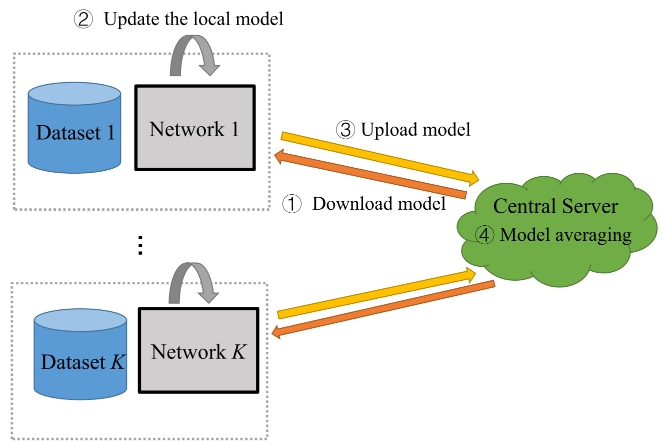

Fig. 1. General FL framework, where a single global model is trained.

In real-world applications, the training data across different clients are non independent and identically distributed (non-IID),[1] which is also known as data heterogeneity [13]–[16]. This occurs because the data sources, namely, the local clients, are often personal devices where the training data are generated in distinct manners. Based on this, for FL applications, such as the design of recommendation systems, learning models from decentralized and heterogeneous training data leads to the following dilemma. On one hand, the training data owned by each client itself may not be enough to learn a satisfactory model, which means that an individual client needs to learn from others. On the other hand, if a client learns from the other clients possessing data with significantly different distributions compared with itself, the test performance at this client may be degraded, which is the case encountered by the FL algorithms that design a single global model [8], [13], [17], [30]. For this case, some researchers have proposed algorithms to take full advantage of the heterogeneous training data under the FL framework [13], [30]. For

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[1]In this article, the training data are assumed to be independent but not identically distributed across different clients, as done in [8].

example, in [13], variance reduction was employed to alleviate the negative impact of client-drift in the presence of heterogeneous data sets. Li *et al.* [30] modified FedAvg [8] and a proximal term was incorporated into the objective function to reduce the adverse influences of data heterogeneity on the convergence stability. However, training a single global model as done in [8], [13], and [30] is intrinsically prone to the negative impacts of heterogeneous training data. In fact, learning a shared global model in FL originates from the learning tasks based on IID training data, whose learning performance deteriorates in the presence of heterogeneous data [8]. As a possible way of resolving the above dilemma caused by data heterogeneity, the framework of clustered FL has been considered in [17], [27]–[29], and [31]–[33]. However, some of these works in [17], [27]–[29], and [31]–[33] assume that the server knows the cluster identity of each client *a priori*. Besides, these prior works partition the clients into nonoverlapping clusters and the training data at each client is only used to update a single cluster model, which leads to inefficient utilization of the information in the decentralized data. Among these works, iterative federated clustering algorithm (IFCA) has been proposed very recently [17], which not only provides convergence guarantees, but also incurs low computational costs at the central server and is thus quite suitable for training large models. IFCA is implemented by alternating between identifying the cluster membership of the clients and optimizing the cluster models, which leads to better learning performance than the global model method. This is because in IFCA, each client learns from only those clients that have similar data distributions. Even though IFCA makes better use of the non-IID training data, it has some drawbacks.

1) It partitions the clients into nonoverlapping clusters and the knowledge of each participating client is exploited by only one cluster during each round, which results in an inefficient utilization of the information contained in the decentralized data.
2) Development of nonoverlapping clusters neglect the fact that different clusters have gradual and blurry boundaries, and are thus not able to characterize the complex nature of the differences and similarities among the clients.

For many traditional problems in various fields of engineering, clustering algorithms have been widely investigated, whose goal is to group similar data into the same cluster [18]–[24]. Due to its simplicity, the hard clustering method, where each data sample is assigned to a single cluster, has attracted much attention [20], [21]. Later, to take the gradual boundaries among different clusters into account, soft clustering methods have been developed [22]–[24], which allow for cluster overlaps and, therefore, are more flexible to reflect the complex nature of the data. To the best of our knowledge, the advantages of soft clustering have not yet been exploited under the FL scenario to deal with data heterogeneity.

In this article, by combining the advantages of IFCA [17] and soft clustering [22]–[24], we propose a new algorithm named FL with soft clustering (FLSC) for FL problems under data heterogeneity across different clients. During each round of FLSC, the participating clients are partitioned into multiple overlapping clusters and each of them transmits the updated model after local model fusion and updating with its private data set. After that, the central server updates the cluster models based on the received models and client identities. Compared with IFCA, FLSC not only improves the utilization of local information by sharing the knowledge of each participating client among multiple clusters, but also utilizes soft clustering to capture the complex nature of the data distributions associated with different clients and therefore, enhances the learning performance of each local client. Experiments on the MNIST [25] and Fashion-MNIST [26] data sets demonstrate the superiority of FLSC compared with IFCA and the global model method.

It is worth noting that FLSC is not a direct extension of IFCA and the differences between FLSC and IFCA lie in the following aspects.

1) In FLSC, each client belongs to more than one cluster and it performs local training based on multiple clusters models after model fusion. In contrast, in IFCA each local client directly updates the best suited model received from the server without any fusion. From this perspective, the clients perform different computation tasks in FLSC and IFCA.
2) For IFCA and FLSC, the cluster models are updated in different manners at the central server based on different clustering methods. It is also worth noting that, although the computation tasks at the local clients and the central server are different in FLSC and IFCA, the former incurs almost no extra communication overhead compared with the latter.

Our main contributions are summarized as follows.

1) By combining the advantages of soft clustering and IFCA in an innovative manner, we propose FLSC to solve the FL problem with heterogeneous training data. FLSC involves novel approaches to conduct the local training at the clients and to perform model fusion at the central server based on the overlapping clustering of the clients.
2) The performance of FLSC is experimentally validated on two popular data sets for image classification tasks under the FL scenario. The results show the potential of FLSC for improving the training performance compared with the baseline methods, i.e., IFCA and the global model method.

The remainder of this article is organized as follows. The considered problem and the background are introduced in Section II. We propose the FLSC algorithm and present its rationale in Section III. In Section IV, experimental results are provided to show the superiority of FLSC over the baseline methods. We provide concluding remarks in Section V at the end of this article.

II. Problem Formulation and Background

A. Problem Formulation

The FL problem studied in this article is formulated as follows. Without loss of generality, image classification tasks under the FL scenario are considered [8]. Suppose there are K clients and a central server in the network. Each client owns a labeled data set $\mathcal{D}_k := \left\{\left(x_i^k, y_i^k\right)\right\}_{i=1}^{N_k}$, $k = 1, \ldots, K$, where x_i^k denotes the ith data sample at the kth client, $y_i^k \in \{1, 2, \ldots, Z\}$ is the label of x_i^k among Z classes, and the kth client possesses N_k data samples. Since data sets at different clients are generated and collected by different end-users in real-world applications, the training data owned by those clients follow distinct distributions P_k, $k = 1, \ldots, K$. In other words, the training data are non-IID across different clients [13]–[16]. The goal of the system is to train C global models sharing the same model architecture and each client eventually obtains the best suited model among the C models for itself [17]. To this end, the central server and the clients communicate with each other under some communication protocols. To protect the data privacy of the clients and reduce the communication overhead, only model parameters are transmitted in the network instead of the raw data sets.

B. Background

For the problem introduced in Section II-A, an existing approach named IFCA has been proposed very recently [17]. In IFCA, during each round, the server broadcasts C cluster models to all the participating clients. Upon receiving the C models, each participating client finds the model therein that attains the lowest local empirical loss to estimate its cluster identity, updates the selected model with its local data set and sends the updated model to the central server together with the client identity. Then, the central server averages the received models that are accompanied with the same identity to update all cluster models. Although the IFCA framework represents a significant advance in the field of FL, it partitions the clients into nonoverlapping clusters, which results in an inefficient utilization of local information. Moreover, IFCA neglects the fact that different clusters have gradual and blurry boundaries and is not able to take full advantage of the complex nature of decentralized training data in the presence of data heterogeneity.

To tackle the above problems in IFCA, we improve it by combining the advantages of soft clustering and IFCA, and propose a new FLSC algorithm to be introduced in Section III.

III. Federated Learning With Soft Clustering

In this section, we first propose the FLSC algorithm and then analyze the difference between FLSC and the baseline methods, including IFCA and the global model method. Based on that, we explain the rationale behind the superiority of the proposed FLSC algorithm.

A. Proposed Algorithm

In this article, we propose a new algorithm named FLSC to deal with the problem formulated in Section II-A. The proposed algorithm is presented as Algorithm 1, and its schematic is shown as Fig. 2. Before the training starts, the kth client randomly selects $N (1 \le N \le C)$ client identities from C clusters as its initial client identities, and the initial identity set containing the client identities of the kth client is denoted as $\mathcal{A}_0^k = \left\{\alpha_0^{k,1}, \alpha_0^{k,2}, \ldots, \alpha_0^{k,N}\right\}$, where $k = 1, \ldots, K$, $\alpha_0^{k,n} \in \{1, 2, \ldots, C\}$, and $n = 1, \ldots, N$. The set containing the parameters of the cluster models is randomly initialized as $\mathcal{W}_0 = \left\{\mathbf{w}_0^c = \mathbf{w}_0, c = 1, \ldots, C\right\}$, where $\mathbf{w}_0^c$ denotes the parameters of the cth cluster model. For the simplicity of notations, hereinafter models are represented by their parameters. For instance, the cluster model set $\mathcal{W}_0$ refers to the

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Algorithm 1: FLSC (The Proposed Algorithm)

Input: initial parameters $\mathbf{w}_0$.
Initialization: The identity sets of the clients are randomly initialized as $\mathcal{A}_0^k = \left\{\alpha_0^{k,1}, \alpha_0^{k,2}, \ldots, \alpha_0^{k,N}\right\}$, where $k = 1, \ldots, K$ and $\alpha_0^{k,n} \in \{1, 2, \ldots, C\}$, $\forall n$. The cluster models are initialized as $\mathcal{W}_t = \left\{\mathbf{w}_0^c = \mathbf{w}_0, c = 1, \ldots, C\right\}$.
For $t = 0, \ldots, T$ **do**
  Randomly select a set $\mathcal{P}_t$ of participating clients.
  The central server do: broadcast $\mathcal{W}_t = \left\{\mathbf{w}_t^c, c = 1, \ldots, C\right\}$.
  If $t > 0$ **do**
    For $k \in \mathcal{P}_t$ **each client in parallel do**

$$\mathcal{A}_t^k \leftarrow \underset{\mathcal{A}=\{\alpha_1,\ldots,\alpha_N\}\subset\{1,\ldots,C\}}{\arg\min} \sum_{n=1}^{N} \mathrm{Loss}\left(\mathbf{w}_t^{\alpha_n}, \mathcal{D}_k\right). \quad (1)$$

    end
  end
  For $k \in \mathcal{P}_t$ **each client in parallel do**
    Local model fusion:

$$\tilde{\mathbf{w}}_t^k \leftarrow \sum_{c\in\mathcal{A}_t^k} \mathbf{w}_t^c \Big/ \left|\mathcal{A}_t^k\right|. \quad (2)$$

    Local model update:

$$\hat{\mathbf{w}}_t^k \leftarrow \mathrm{SGD}_{B_1,E}\left(\tilde{\mathbf{w}}_t^k, \mathcal{D}_k\right). \quad (3)$$

    The k-th client transmits $\hat{\mathbf{w}}_t^k$ and $\mathcal{A}_t^k$ to the server.
  end
  The central server do:

$$\mathcal{W}_{t+1} \leftarrow \mathrm{UpdateClusters}\left(\left\{\hat{\mathbf{w}}_t^k, \mathcal{A}_t^k, k \in \mathcal{P}_t\right\}\right). \quad (4)$$

  end
end
***Until*: convergence**
Function: $\underline{\mathrm{UpdateClusters}\left(\left\{\hat{\mathbf{w}}_t^k, \mathcal{A}_t^k, k \in \mathcal{P}_t\right\}\right)}$ at the central server:
Define C empty sets $\mathcal{H}_c$, $c = 1, \ldots, C$.
For $c = 1, \ldots, C$ **do**
  For $k \in \mathcal{P}_t$ **do**
    If c **is in** $\mathcal{A}_t^k$ **do**
      Add $\hat{\mathbf{w}}_t^k$ to set $\mathcal{H}_c$.
    end
  end
end
For $c = 1, \ldots, C$ **do**
  $\mathbf{w}_{t+1}^c \leftarrow \sum_{\hat{\mathbf{w}}_t^k\in\mathcal{H}_c} \frac{|\mathcal{D}_k|}{\sum_{\hat{\mathbf{w}}_t^i\in\mathcal{H}_c}|\mathcal{D}_i|} \hat{\mathbf{w}}_t^k$.
end
return $\mathcal{W}_{t+1} \leftarrow \left\{\mathbf{w}_{t+1}^c, c = 1, \ldots, C\right\}$

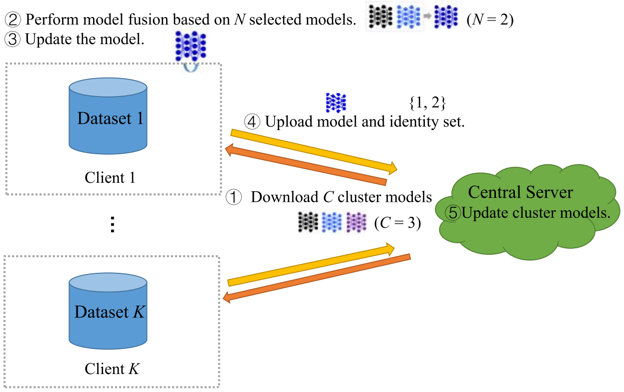

Fig. 2. Schematic of the proposed FLSC algorithm ($C = 3, N = 2$).

set containing the parameters of all cluster models. Similarly, model $\mathbf{w}_0$ refers to the model with parameters $\mathbf{w}_0$.

In the tth round ($t \geq 0$), a subset $\mathcal{P}_t$ of clients are randomly selected to be participating clients, where $|\mathcal{P}_t| = P$ and $|\mathcal{P}_t|$ is the cardinality of set $\mathcal{P}_t$. Then, the central server broadcasts the cluster models in $\mathcal{W}_t = \{\mathbf{w}_t^c, c = 1, \ldots, C\}$ to all the participating clients. Upon receiving those cluster models, each participating client determines its client identities and updates its identity set $\mathcal{A}_t^k$ which contains N indices of the clusters that the client belongs to,[2] $k \in \mathcal{P}_t$. For $t = 0$, the initial identity set $\mathcal{A}_0^k$ is adopted by the clients. In the tth round ($t > 0$), each participating client computes the empirical loss of each cluster model using its local data set and obtains its identity set $\mathcal{A}_t^k = \{\alpha_t^{k,1}, \alpha_t^{k,2}, \ldots, \alpha_t^{k,N}\}$ according to (1) in Algorithm 1, which contains the indices of N cluster models attaining the lowest loss compared with others. Without loss of generality, it is assumed that the model $\mathbf{w}_t^{\alpha_t^{k,1}}$ attains the lowest loss among the N selected cluster models at the kth client, $k \in \mathcal{P}_t$. The loss function in (1) depends on the specific learning tasks. For the considered image classification tasks, the loss function is the cross entropy loss given as

$$\text{Loss}(\mathbf{w}, \mathcal{D}) \triangleq -\sum_{(x,y)\in\mathcal{D}} \mathbf{h}^T(y) \log \mathbf{p}(\mathbf{w}, x) \tag{5}$$

where $\mathcal{D}$ is the data set, $\mathbf{p}(\mathbf{w}, x)$ is the soft prediction of model $\mathbf{w}$ based on the data sample x, and $y \in \{1, 2, \ldots, Z\}$ is the true label of x. In (5), $\mathbf{h}(y)$ denotes the $Z \times 1$ one-hot vector with all elements being zero and the yth element being one and $\mathbf{h}^T$ is the transpose vector of $\mathbf{h}$.

Next, by performing model averaging as shown in (2), each participating client fuses $\{\mathbf{w}_t^{\alpha_t^{k,1}}, \mathbf{w}_t^{\alpha_t^{k,2}}, \ldots, \mathbf{w}_t^{\alpha_t^{k,N}}\}$, i.e., the cluster models from the clusters it belongs to, and obtains model $\tilde{\mathbf{w}}_t^k$, $k \in \mathcal{P}_t$. After that, each participating client updates the resulting model $\tilde{\mathbf{w}}_t^k$ by performing E training passes over $\mathcal{D}_k$, i.e., the private data set of the kth client, using stochastic gradient descent (SGD) with mini-batch size B_1 as shown in (3), $k \in \mathcal{P}_t$. Then each participating client transmits both the locally updated model $\hat{\mathbf{w}}_t^k$ and the identity set $\mathcal{A}_t^k$ to the central server, $k \in \mathcal{P}_t$. Upon receiving the information from all the participating clients, the server updates the cluster models using (4), where the UpdateCluster function is provided at the bottom of Algorithm 1. To be more specific, the UpdateCluster function at the central server aims at performing model weighted averaging on C groups of models to update C cluster models, respectively, where the cth group of models contains all the models received from the clients in the cth cluster, $c = 1, \ldots, C$, and the weights are determined by the number of training data samples owned by the clients. Note that, different from IFCA, since each client belongs to N clusters, each model received by the central server is used to update N cluster models at the server in each round. Next, the difference between the baseline methods and the proposed FLSC algorithm will be stated in more details.

[2]Without loss of generality, in this article, we assume that each client belongs to N clusters. Our results can be easily extended to the case where different clients belong to different number of clusters.

B. Difference Between FLSC and the Baseline Methods

Through comparison between FLSC given as Algorithm 1 and IFCA given in [17], it can be seen that the structures of IFCA and FLSC with $1 < N < C$ are different from each other mainly in two aspects.

1) In FLSC, each client belongs to N clusters and it performs local training in (3) after fusing multiple cluster models as shown in (2). Different from this, in IFCA each local client directly updates the best suited model among all the models received from the server without model fusion.
2) For IFCA and FLSC, clustering methods are different and the cluster models at the central server are updated in different manners, which can be observed from the UpdateCluster function in Algorithm 1 and the model averaging function in IFCA [17]. To be more specific, each model sent by the clients is used to update one cluster model at the central server in IFCA while each received model at the server is utilized to update multiple cluster models in the proposed FLSC algorithm.

It is worth noting that, FLSC and IFCA incur almost the same communication overhead per round. To be more specific, for both algorithms, the communication cost in each round is induced by the transmission of the cluster models from the server to the participating clients and uploading one model from each participating client to the server together with the cluster indices. Although a handful of model indices are transmitted by each participating client in FLSC while a single model index is sent by each participating client in IFCA, the communication cost induced by the transmission of model indices is negligible compared with that incurred by sending a large set of model parameters. From this perspective, the communication overhead of FLSC and IFCA is almost the same. Besides, compared with the computational cost induced by gradient descent for the model update in FLSC and IFCA, the computational cost of model averaging at the local clients in FLSC is negligible, which indicates that the computation complexity of FLSC and IFCA are approximately the same. Another baseline method is the global model method shown in Fig. 1, which trains a single global

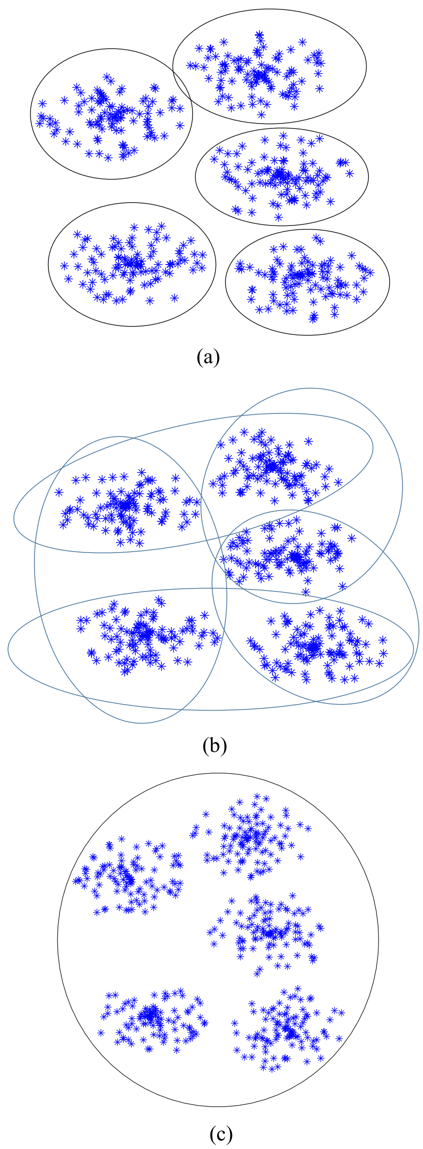

Fig. 3. Schematics of the three algorithms. (a) IFCA. (b) FLSC (The proposed, $C = 5, N = 2$). (c) Global model method.

model in the classical manner [8] without partitioning the clients into clusters and training different models for each cluster.

In fact, IFCA [17] and the global model method [8] can be regarded as two special cases of FLSC. By setting $N = 1$, FLSC is equivalent to IFCA; and by setting $N = C$, FLSC attains the same learning performance as the global model method. Actually, the hyperparameter N $(1 < N < C)$ controls the degree of overlap of the clusters in FLSC and a larger value of N indicates highly overlapping clusters. Based on that, hereinafter we refer to N as the *degree of overlap* in FLSC.

C. Rationale Behind the FLSC Algorithm

Next, the rationale behind the superiority of the proposed FLSC will be discussed via comparison among FLSC, IFCA, and the global model method.

The schematics of the three algorithms are presented in Fig. 3, where each blue point represents a client and each circle represents a cluster. In IFCA shown in Fig. 3(a), each client only belongs to a single cluster and the clusters are pairwise disjoint, which means that the information of each client can be utilized solely within a cluster. In FLSC shown in Fig. 3(b), each client belongs to multiple clusters, which can participate in the model training in those clusters simultaneously. From this perspective, the local information in the decentralized training data can be utilized much more efficiently thanks to overlapping clustering in FLSC. We can also observe from Fig. 3(b) that overlapping clusters capture the similarities and differences among different clients in a more appropriate way, which contributes to better learning performance in each cluster. In the global model method as shown in Fig. 3(c), all the clients are incorporated into a single cluster. Since the data distributions on some clients are significantly different from others, it is difficult for the global model method to attain satisfactory personalized learning performance for all the clients.

From the above analysis, the values of N that are either too large, e.g., $N = C$, or too small, e.g., $N = 1$, result in degraded learning performance of FLSC. In other words, when implementing FLSC, an appropriate value of N needs to be selected to guarantee optimal learning performance. This will be verified by experiments in Section IV, where some guidance about how to determine the default value of N are provided based on the experimental results.

IV. Experiments

In this section, we provide experimental validation of the proposed FLSC algorithm on the MNIST [25] and Fashion-MNIST [26] data sets, where image classification tasks are considered under the FL scenario. Comparison with the baseline methods, including IFCA [17] and the global model method [8] is performed to verify the superiority of FLSC.

A. Data Sets

The data sets utilized in the experiments are introduced as follows.

MNIST [25] is composed of 70k images at a resolution of 28×28, which covers ten handwritten digits from 0 to 9. It has a total number of 60k training images and 10k test images. Only the training images are used in our experiments.

Fashion-MNIST [26] consists of 60k images covering ten classes of fashion items at a resolution of 28×28, which includes 60k training images and 10k test images. We only use the training images in the experiments.

B. Baseline Methods

Two state-of-the-art methods are used as the baseline in the experiments, which are introduced as follows.

In *IFCA*, [17], all the clients learn their cluster identities and train the cluster models alternatively, where the clusters are nonoverlapping. This method can be implemented by setting $N = 1$ in Algorithm 1.

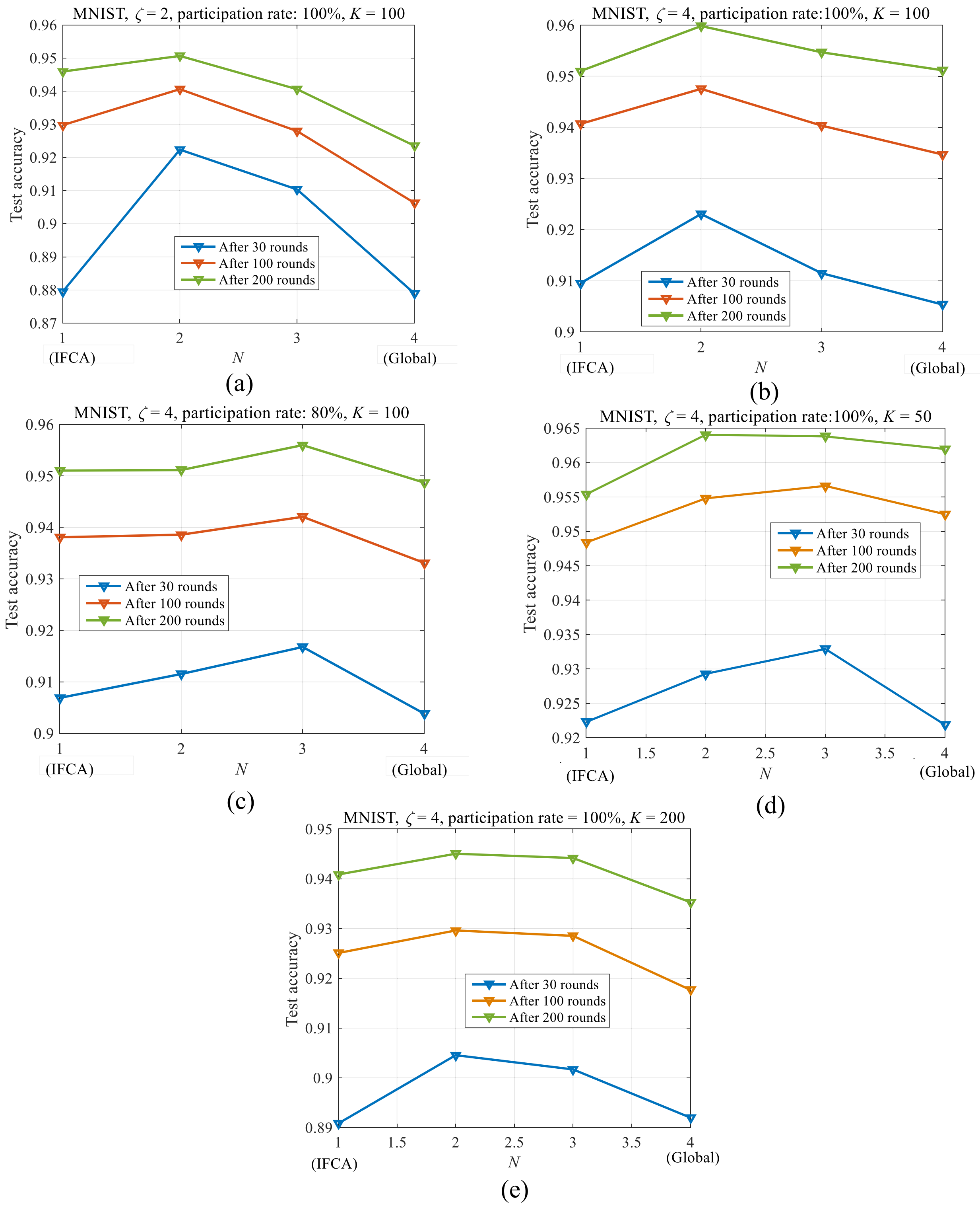

Fig. 4. Overall test accuracy versus the degree of overlap N on the MNIST data set using MLP after finishing a certain number of rounds of training under different degrees of data heterogeneity, different participation rates, and different number of clients. (a) $\xi = 2, K = 100$, participation rate is 100%. (b) $\xi = 4, K = 100$, participation rate is 100%. (c) $\xi = 4, K = 100$, participation rate is 80%. (d) $\xi = 4, K = 50$, participation rate is 100%. (e) $\xi = 4, K = 200$, participation rate is 100%.

In the *global model method* [8], all the clients learn a single global model collectively. All the clients share the same cluster identities in the global model method, which can be implemented by setting $N = C$ in Algorithm 1.

C. Implementation Details

In these experiments, we adopt two types of neural networks, i.e., multilayer perceptrons (MLP) and convolution neural networks (CNNs). The number of clients is denoted by

K and the training images are divided into K parts, each of which contains the same number of data samples. To simulate the non-IID data in real-world applications, most of the clients are assigned data samples of ζ classes. The specific values of ζ in the experiments are provided in Section IV-D. Note that the value of ζ indicates the degree of heterogeneity of the training data sets. In other words, a smaller value of ζ indicates a higher degree of data heterogeneity. This non-IID data partition setting can be realized by arranging the training images according to their labels and then dividing them into $K\zeta$ equal-size segments [8]. After that, each client obtains ζ segments of data samples. For each client, 80% of the data samples are used to train models, which we refer to as *local training data*, and the rest 20% of them are used to test the classification accuracy of the trained models, which we refer to as *local test data*.

The proposed method and the baseline method are implemented according to Algorithm 1 and the total number of clusters is fixed at $C = 4$ unless specified, as done in [17] and [28]. As pointed out in [28], $C = 4$ is a good choice for the number of clusters because larger values of C induce heavier communication burden and computation burden at the local clients while smaller values of C are not able to attain satisfactory learning performance under the framework of FL with clustered clients. After training for a certain number of rounds, each local client selects the cluster model with the lowest empirical loss on its *local training data* and evaluates the classification accuracy of the selected model on its *local test data*. Then the *overall test accuracy* is recorded as the averaged result of the classification accuracy at the local clients. Under each setting, we run the experiments independently three times so as to alleviate the effects of randomness in the training.

D. Experimental Results and Discussion

In the first experiment, for FLSC, IFCA, and the global model method, MLP models are trained on the MNIST data set for different values of ζ, different number of clients in the network, and different participation rates, i.e., the ratio of the number of participating clients per round to the total number of clients. In this experiment, each participating client performs $E = 10$ training passes over its local training data during each round and an SGD optimizer with learning rate $= 0.005$, mini batchsize $= 50$, and momentum $= 0.5$ is adopted. In Fig. 4(a)–(e), we depict the overall test accuracy versus the degree of overlap N after finishing a certain number of rounds during the training process, where the test accuracy attained by IFCA is shown at the points corresponding to $N = 1$ and that of the global model method is shown at the points $N = C = 4$. From Fig. 4, it can be observed that, under most of the experimental settings, the optimal learning performance of FLSC is attained at $N = 3$. Although under some cases FLSC achieves the optimal performance at $N = 2$, it is worth noting that FLSC outperforms IFCA and the global model method both when $N = 2$ and $N = 3$. Based on these results, the efficacy of FLSC for different values of N is verified and we suggest setting the default value of N to be 3. In

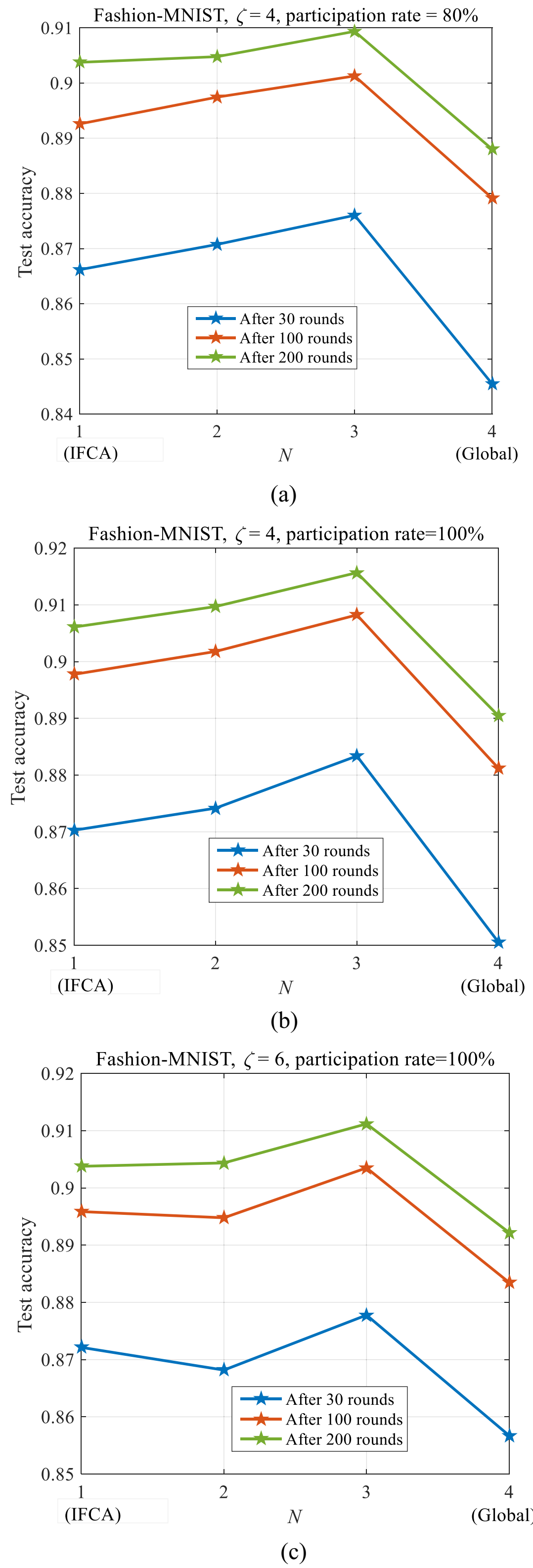

Fig. 5. Overall test accuracy versus the degree of overlap N on the Fashion-MNIST data set using CNN after finishing a certain number of rounds of training under different degrees of data heterogeneity and different participation rates. (a) $\xi = 4$, participation rate is 80%. (b) $\xi = 4$, participation rate is 100%. (c) $\xi = 6$, participation rate is 100%.

Fig. 6(a), we plot the overall test accuracy versus the number of rounds when $\xi = 4, K = 100$, and the participation rate is 100% for different algorithms, where $N = 3$ is used in the

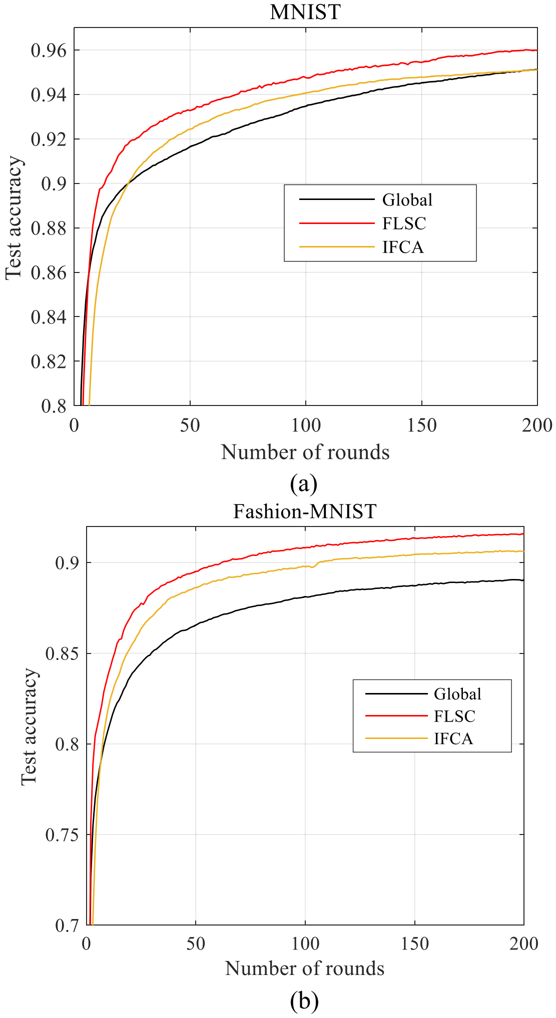

Fig. 6. Overall test accuracy versus the number of rounds attained by different algorithms (The global method, IFCA, and the proposed FLSC) on the MNIST and Fashion-MNIST data sets, where $\xi = 4, K = 100$ and the participation rate is 100%. (a) Results on the MNIST data set. (b) Results on the Fashion-MNIST data set.

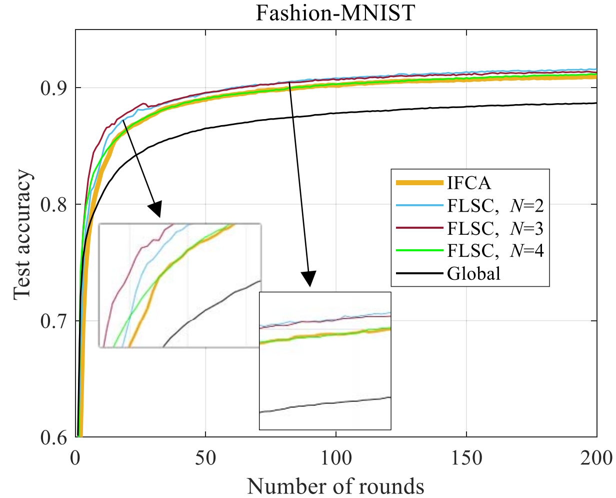

Fig. 7. Overall test accuracy versus the number of rounds attained by different algorithms (The global method, IFCA, and the proposed FLSC) on the Fashion-MNIST data sets, where $\xi = 4, K = 100, C = 5$, the participation rate is 100 %, and FLSC is implemented under different values of N.

proposed FLSC. The convergence of FLSC is verified by the experimental results in Fig. 6(a). Besides, it shows that FLSC converges faster to a higher value of test accuracy compared with the baseline methods. The superiority of FLSC results from the fact that the clients are partitioned into overlapping clusters. By doing this, FLSC utilizes the local information of the clients more efficiently and characterizes the similarities among the clients more accurately, which makes the cluster models converge faster in the direction of the steepest descent.

In the second experiment, for FLSC, IFCA, and the global model method, CNN models are trained on the Fashion-MNIST data set with different values of ζ and different participation rates, where the number of clients is fixed at $K = 100$ and each participating client performs $E = 10$ training passes over the local training data during each round using an SGD optimizer with learning rate = 0.001, mini batch-size = 50, and momentum = 0.5. In Fig. 5(a)–(c), the overall test accuracy versus the degree of overlap N after finishing a certain number of rounds is plotted. Similar to the results of the first experiment, the test accuracy attained by IFCA and the global model method is shown at the points, where $N = 1$ and $N = C = 4$, respectively. It can be easily seen from Fig. 5 that FLSC attains the optimal learning performance with $N = 3$ and it outperforms the baseline methods, which verifies the effectiveness of setting the default value of N to be 3. Similar to the experimental results in the first experiment, FLSC outperforms IFCA and the global model method both when $N = 2$ and $N = 3$, which demonstrates its efficacy. To further demonstrate the efficacy of setting the default value of N to be 3, in Fig. 7, we plot the overall test accuracy attained by FLSC under different values of N and the baseline methods versus the number of rounds when the total number of clusters is $C = 5$, the participation rate is 100%, $\xi = 4$, and $K = 100$. It can be easily seen from Fig. 7 that the optimal learning performance is achieved at $N = 3$ when FLSC is implemented, which outperforms the IFCA and the global model method. Based on that, it is reasonable to set the default value of N to be 3. In addition, in Fig. 6(b), the overall test accuracy versus the number of rounds is depicted when $\xi = 4, K = 100$, and the participation rate is 100% for different algorithms, where the value $N = 3$ is selected for the implementation of FLSC. The result in Fig. 6(b) further demonstrates the superior convergence property of FLSC compared with the other two baseline methods.

V. Conclusion

In this article, the problem of FL with non-IID training data was considered. To overcome the shortcomings of the state-of-the-art IFCA method, we proposed a new FLSC algorithm by combining the strengths of IFCA and soft clustering, where clients are partitioned into overlapping clusters. In this way, the information at each client can be utilized more efficiently by multiple clusters at the same time and the similarities among the clients can be characterized more appropriately to enhance the learning performance. We ran experiments on the MNIST and Fashion-MNIST data sets, which demonstrated that FLSC attains better learning performance compared with the baseline methods, namely, the global model method and IFCA. In the future, we plan to conduct explicit theoretical analysis on the convergence criterion and convergence performance of FLSC and to investigate the extension of the proposed FLSC algorithm which is robust to the Byzantine attacks in the FL system [32]. We also plan to extend the proposed method to more general cases where the training data are dependent across different clients using methods, such as provided in [37].

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